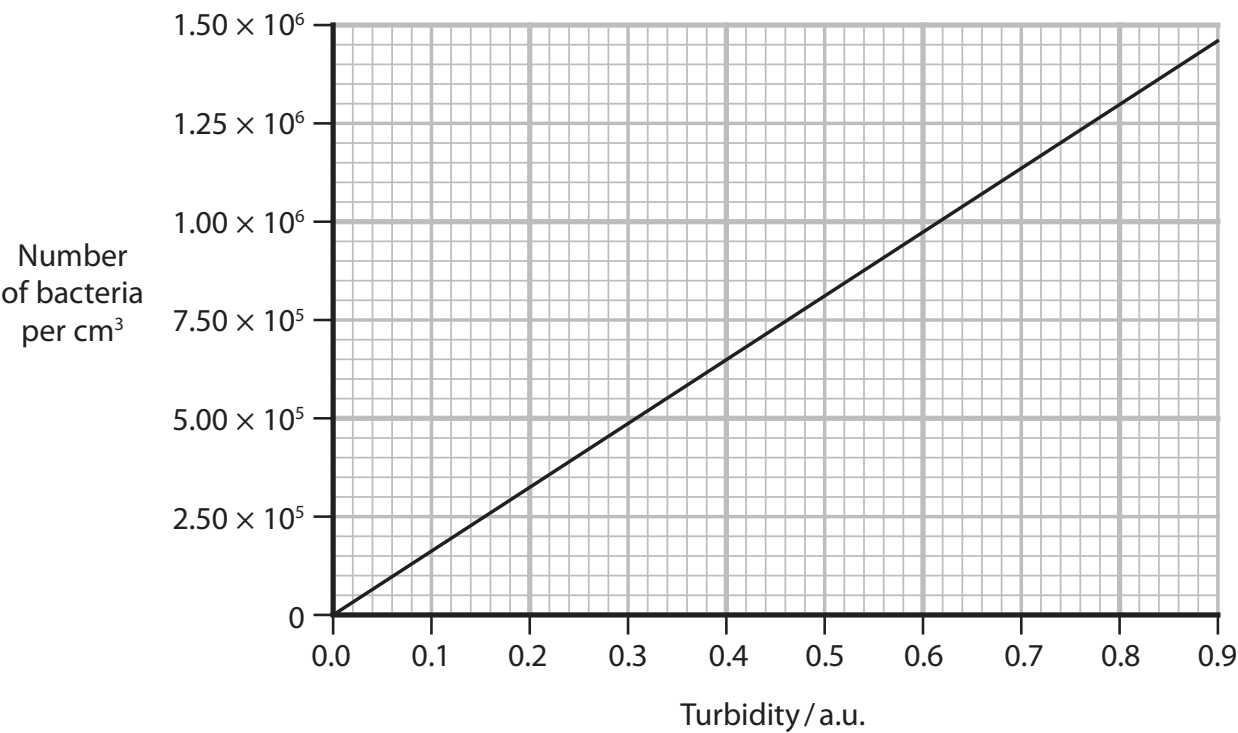


3 The growth of bacteria in a liquid culture was investigated over a 14 hour period, using an optical method (turbidity).

A calibration curve was used to convert the turbidity readings into the number of bacteria per cm³ of liquid culture.

The graph shows the calibration curve used.



The table shows the results for the first 10 hours of culture.

Time of culture / hours	Turbidity / a.u.	Increase in cell number in each two hour period
0	0.01	
2	0.01	0
4	0.04	5.0 × 10 ⁴
6	0.56	
8	0.80	4.0 × 10 ⁵
10	0.80	0

(a) Complete the table to show the increase in cell number between 4 and 6 hours of culture.

(b) Explain the changes during the first six hours of this culture.

(2)

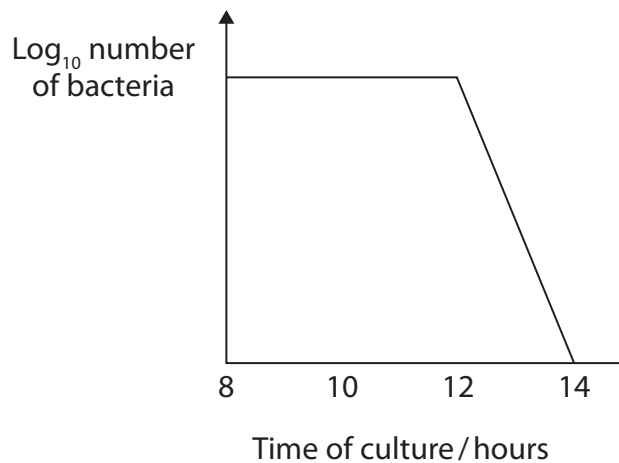
(c) The number of cells remained constant after 8 hours to the end of the 14 hour period of this culture.

Bacterial growth was also investigated using dilution plating and colony counts.

(i) Describe the method used in dilution plating.

(2)

- (ii) The sketch shows the shape of the growth curve produced using dilution plating and colony counts between 8 and 14 hours of this culture.



There are differences in the numbers of bacteria between 8 and 14 hours measured using optical methods (turbidity) and measured using dilution plating and colony counts.

Explain the differences obtained by these two methods.

(3)

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(Total for Question 3 = 8 marks)